Week 10, Lecture 1: Likelihood Ratio Tests, AIC, and Regularization

Instructor notes. Part 1: prep reading. Part 2: podium cheat sheet

How to prepare (for students)

- **Big picture.** We have spent several weeks fitting models (FIT): likelihood, optimization, MCMC. This week moves to COMPARE: when several models could explain the same data, how do we decide which ones the data support? Monday covers likelihood ratio tests, AIC, and an alternative to choosing among models at all: fitting one big model with a penalty that shrinks unneeded coefficients toward zero (regularization). Wednesday covers the Bayesian versions.
- **Read:** Bolker 6.6 through 6.6.2 (printed p. 267-280) and the paragraph on AIC weights on p. 284. Also skim Tredennick et al. (2021), *A practical guide to selecting models for exploration, inference, and prediction in ecology* (Ecology 102: e03336); focus on why the purpose of an analysis should decide how models are compared.
- **Review:** the likelihood ratio interval and the 1.92 cutoff from Week 7; the negative binomial distribution and the power-law function ax^b from the bestiary; `glm()` with a Poisson family.
- **Tuesday's lab** applies these tools to a larger version of Lab 7's chytrid pond survey, so bring questions.

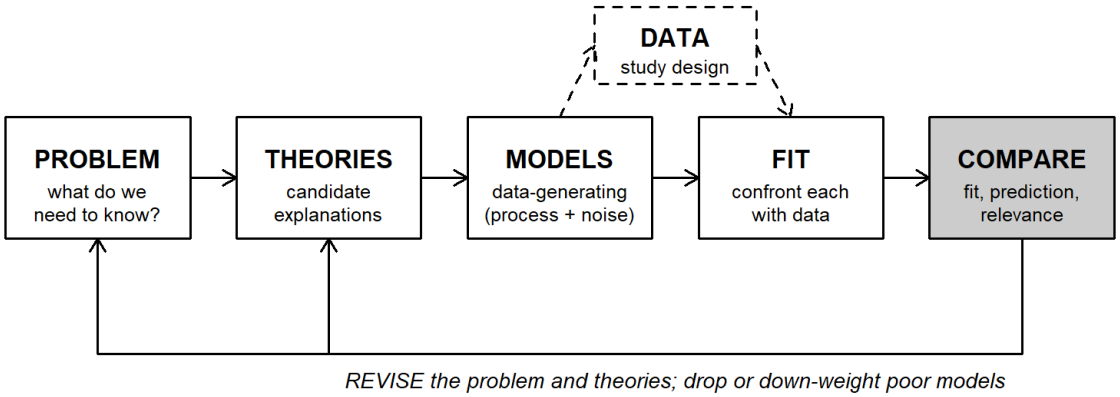
Overview (instructor only)

Beat	Mode	Time
1. Big picture	BOARD	~4 min
2. WOD: *Connect the Dots*	handout	~8 min
3. Likelihood ratio test	BOARD → R	~8 min
4. AIC	BOARD → R	~10 min
5. Ridge and lasso	BOARD → R	~12 min
6. Wrap and discussion	BOARD	~5 min

- R = Monday section of `CODE_Week10_ModelComparison.R`. No Stan debrief (lab).
- **Behind?** Skip the Poisson pause and AICc/BIC; past minute ~38, do section 5's board part only and open Wednesday with the glmnet demo.

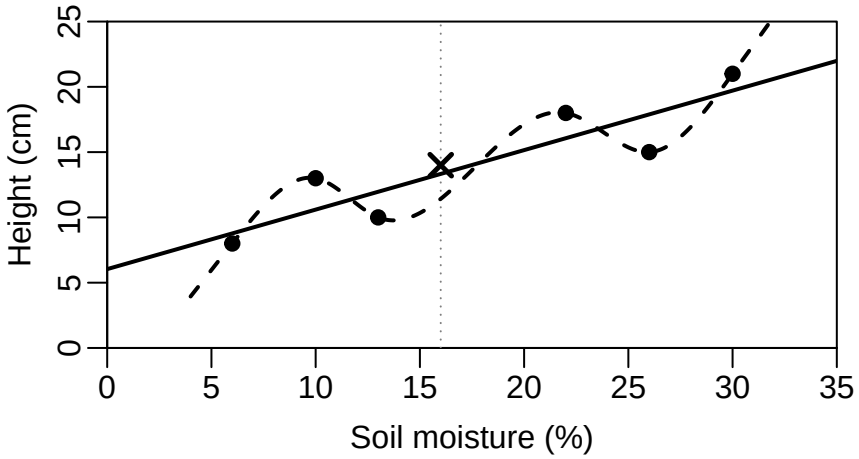
Part 1: Lecture content

1. Big picture (~4 min) [BOARD]



- COMPARE = fit / prediction and rivals / relevance; today is the middle one
- Tools: penalize in-sample fit (LRT, AIC), held-out data (CV), or fit skeptically (ridge, lasso)
- Survivors are corroborated, not proven

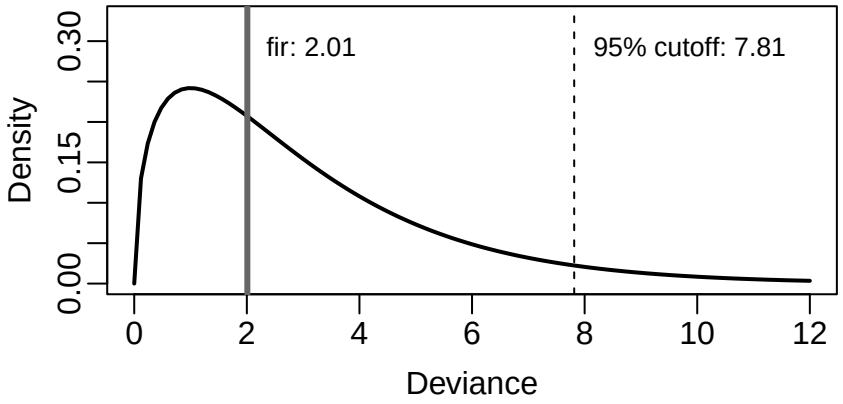
2. WOD reveal (~8 min) [handout]



- Residual SS: line 24.0, curve 0
- New plot **14 cm**; line 13.3, curve 11.4
- Land: in-sample fit always improves with flexibility

3. Likelihood ratio test (~8 min) [BOARD → R]

- Fir (Bolker 6.6): cones $\sim$ NegBin($a\,\mathrm{DBH}^b, k$); full (wave/non-wave a, b, k; 6 params) vs. no-wave (3)
- $D = 2(\mathrm{NLL}_{red} - \mathrm{NLL}_{full}) \sim \chi^2_r$; nested models only
- Week 7's 1.92 = this test with $r = 1$



- **Result:** $D = 2.01$, $p = 0.57$; keep no-wave
- **Pause:** Poisson nested in NB? Yes ($k \to \infty$; legacy LECTURE8 is wrong), but on the boundary, so naive p is ~2x too big. $D = 4{,}052$ here
- **Sticking point:** not rejecting $\neq$ no difference

4. AIC (~10 min) [BOARD → R]

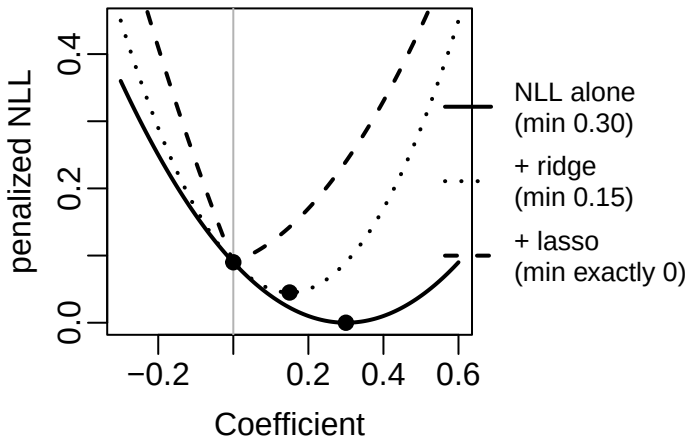
- $\mathrm{AIC} = -2\log\mathcal{L}(\hat{\theta}) + 2k$; nested or not, same response data
- $2k$ = average in-sample optimism (`NOTES_Reference_AICPenalty.Rmd`; don't derive)
- $w_i = e^{-\Delta_i/2} / \sum_j e^{-\Delta_j/2}$; BIC: $k \log n$

	k	NLL	AIC	dAIC	weight	dBIC
no_wave	3	1136.02	2278.03	0.00	0.52	0.00
const_ak	4	1135.76	2279.52	1.48	0.25	4.97
const_b	5	1135.13	2280.27	2.24	0.17	9.21
full	6	1135.01	2282.02	3.99	0.07	14.46
poisson	2	3161.83	6327.66	4049.63	0.00	4046.14

- **Pause:** weight 0.52 = P(true model)? No; relative support within the set (Bolker p. 284)
- **Sticking point:** `const_ak` adds 1 param, NLL down only 0.26, yet $\Delta = 1.48$. A useless parameter costs ≤ 2 (Arnold 2010)

5. Ridge and lasso (~12 min) [BOARD → R]

- Creepers: 50 stations, 10 covariates, simulated (say so); truth = big trees, snags, road distance
- $2^{10} = 1024$ subsets; instead penalize coefficient *size*: NLL+$\lambda \sum \beta_j^2$ (ridge), NLL+$\lambda \sum |\beta_j|$ (lasso)
- Standardize first; λ by 10-fold CV (`lambda.1se`)
- Lasso's corner at 0 gives exact zeros:



- **Full MLE:** all 10 nonzero; north 0.22, elev 0.11 (truth 0)
- **All-subsets AIC:** big_trees, snags, north, road_dist; 8 within 2; best weight 0.08
- **Lasso (CV):** big_trees, snags, road_dist; big trees 0.37 (truth 0.5), road 0.07 (0.3)
- **40 simulated datasets**, exact true set: AIC 12, lasso 13
- **Pause:** road 0.07 vs. 0.3, broken? No: bias traded for lower variance; for effect sizes refit or go Bayesian (Wed.)
- **Sticking point:** neither method is a truth detector

6. Wrap and discussion (~5 min) [BOARD]

- Count penalty / held-out data / size penalty. Tomorrow: pond survey. Wed.: WAIC, LOO, shrinkage priors

Q1. AIC kept northness (8 models within 2, best weight 0.08); the lasso dropped it. Which covariates would you report as "important," and how would you word it?

Q2. In a real survey, the lasso sets grazing to exactly 0. A manager concludes expanding grazing won't affect creepers. Agree?

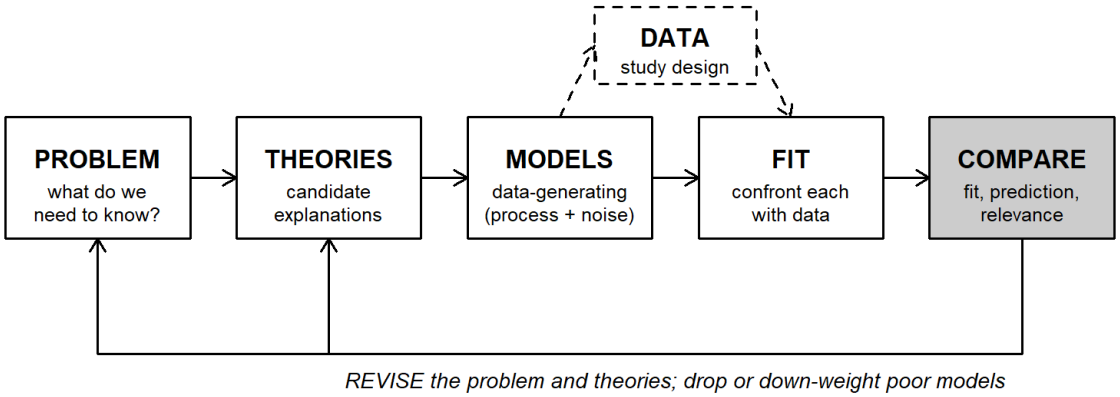
- Q1 lands on: report selection uncertainty; agreed covariates (big trees, snags) are safest
- Q2 lands on: zero = not needed for prediction here, not "no effect"; grazing may act via shrub cover; causal question = COMPARE (3), relevance

PART 2: PODIUM CHEAT SHEET

Terse on purpose. Detail: Part 1. R = Monday section of CODE_Week10_ModelComparison.R.

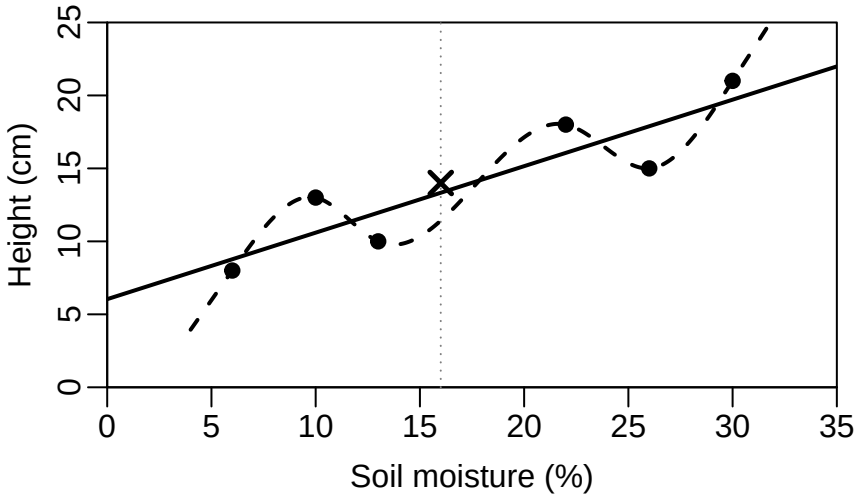
Big picture → WOD → lecture. No Stan debrief (lab). Behind? Skip Poisson pause, AICc/BIC; glmnet demo → Wed.

1. Big picture (~4 min) [BOARD]



- Point at **COMPARE**: fit / prediction + rivals / relevance
- Weeks 6-9 = FIT; today = rivals
- Penalize in-sample fit (LRT, AIC); new data (CV); fit skeptically (ridge, lasso)
- Corroborated, not proven
- Hand out WOD

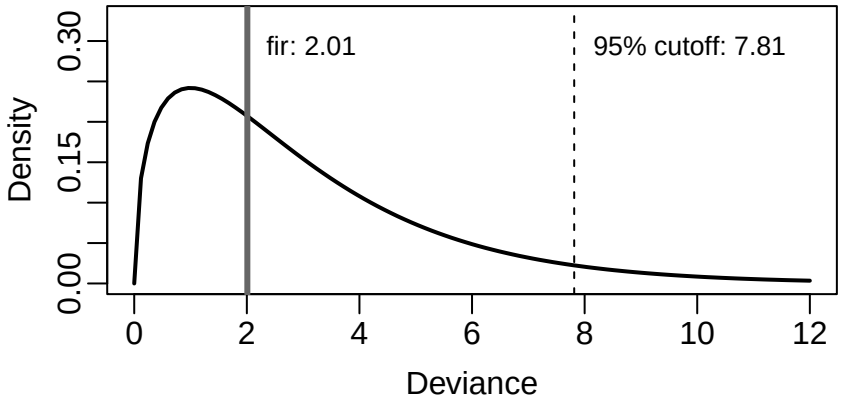
2. WOD reveal (~8 min)



- Residual SS: line 24.0, curve 0
- New plot: **14 cm**; line 13.3, curve 11.4
- Curves disagree in the gap; lines don't
- In-sample fit always improves with flexibility

3. LRT (~8 min) [BOARD → R]

- Fir: NegBin($a\,\mathrm{DBH}^b$, k); wave vs. non-wave
- Full 6 params vs. no-wave 3; **nested**
- $D = 2(\mathrm{NLL}_{red} - \mathrm{NLL}_{full}) \sim \chi^2_r$
- Week 7's $1.92 = 3.84/2$, $r = 1$



- **R:** $D = 2.01$, cutoff 7.81, $p = 0.57$; keep no-wave
- **Pause (skippable):** Poisson nested in NB? Yes ($k \to \infty$), but on the boundary: naive p ~2x too big. $D = 4{,}052$
- Not rejecting $\neq$ no difference

4. AIC (~10 min) [BOARD → R]

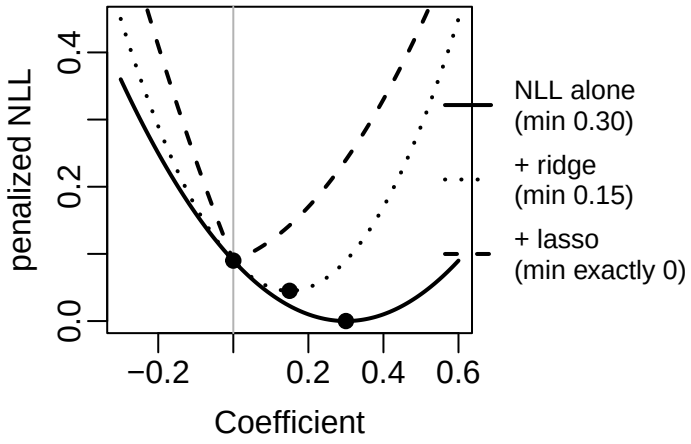
- AIC $= -2 \log \mathcal{L}(\hat{\theta}) + 2k$; any models, same response data
- $2k$ = average optimism (the WOD in one number)
- $w_i = e^{-\Delta_i/2} / \sum_j e^{-\Delta_j/2}$
- AICc: small n; BIC: $k \log n$

	k	NLL	AIC	dAIC	weight	dBIC
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- **Pause:** weight 0.52 = P(true)? No (Bolker p. 284)
- `const_ak`: +1 param, NLL down 0.26, Δ 1.48. Useless param costs ≤ 2

5. Ridge and lasso (~12 min) [BOARD → R]

- Creepers: 50 stations, 10 covariates; simulated; truth = big trees, snags, road
- $2^{10} = 1024$ subsets
- NLL + $\lambda \sum \beta_j^2$ (ridge); NLL + $\lambda \sum |\beta_j|$ (lasso)
- AIC: how many; these: how big. Standardize first



- **R:** MLE: north 0.22, elev 0.11
- Best subset: big_trees, snags, north, road_dist; 8 within 2; weight 0.08
- Paths; CV → `lambda.1se`: big_trees, snags, road_dist
- **Pause:** road 0.07 vs. 0.3. Broken? No: bias for lower variance
- 40 datasets: AIC right 12, lasso 13

6. Wrap and discussion (~5 min)

- Count penalty / held-out data / size penalty
- Tomorrow: pond survey. Wed: WAIC, LOO, shrinkage priors

Q1. AIC kept northness (8 within 2, best weight 0.08); lasso dropped it. What do you report as "important"?

- Report uncertainty; agreed covariates are safest

Q2. Real survey: lasso zeroes grazing. "Expanding grazing won't affect creepers." Agree?

- Zero = not needed for prediction here; not "no effect"
- Grazing via shrub; small effect, 50 stations; causal question = COMPARE (3)